

A learning and motivational dilemma because of Gen AI

Situation

In many areas Gen AI produces outputs of high quality that are good enough “to do the job” of

- passing the bar exam, medical school tests, advanced placement tests in many disciplines etc. in the upper quartile of test results
- writing major sections of a thesis
- coding solutions to complex problems and even finding new algorithms (e.g., AlphaTensor)
- solving math problems
- creating attractive designs
- writing well-argued essays with references

Using AI tools to identify whether AI has been used in generating these outputs does not produce robust results – many false positives and false negatives to be expected given the nature of these tools.

Complication

While the Gen AI may produce a “good enough” output for a job to be done, there remains the question how to judge and appreciate what is “good enough” and what may be missing for excellence.

Achieving this judgment capability rests on a large investment of time, curiosity, and passion to gain the necessary experience and practice – Analogy: 10.000h rule for excellent musicians and athletes according to Malcom Gladwell.

Dilemma

What will motivate students and researchers to invest adequately into the foundations necessary for judgement and excellence if Gen AI outperforms on conventional output-oriented testing and learning methods?

This dilemma is more acute because of the emphasis on metrics rather than interesting insights in modern society, management, and often also in the sciences – see Steffen Mau, *Das metrische Wir*, 2017.

Closing hypothesis

We need to turn our education process upside down by fully embracing the Gen AI tools in the learning journey – e.g., hackathons of prompt engineering or agent-simulations followed by collective reflections on the limitations of the outputs instead of conventional lectures and exercises.

I foresee major resistance to such a paradigm shift especially in humanities and social sciences.